

Restaurant Management System - Software Documentation

1. Project Information

Project Title: Restaurant Management System

Developer: Syed Saad Raza

Programming Language: C++ with Qt Framework

Database: MySQL

Development Tools: Qt Creator, MySQL Connector, Qt Widgets

Operating System: Windows

2. System Overview

In many restaurants, sales, profit, and expense records are traditionally stored on paper, which makes it difficult to manage, track, and analyze business data efficiently. The Restaurant Management System is designed to digitize restaurant operations by providing a centralized system for managing orders, staff, menu items, and sales reports. The system allows management to monitor business performance, track orders from the kitchen, and maintain historical data for long-term analysis.

3. System Users

Admin

Order Taker (Dine-In and Takeaway)

Owner

Kitchen Staff

4. User Roles and Permissions

Admin:

- Add staff members
- Update staff details

- Remove staff members
- Add menu items
- Update menu items
- Remove menu items
- Add restaurant tables

Order Taker:

- Take orders using Takeaway or Dine-In windows
- Assign tables for dine-in orders
- View current orders until they are marked as paid
- Print customer bills

Owner:

- View all staff details
- View all menu details
- View sales reports and profit analysis
- Remove any order if necessary

Kitchen Staff:

- View incoming orders
- Update order status to 'In Progress' or 'Ready'
- Trigger notifications for ready orders

5. Main Features

- Login system with role-based access
- Staff management
- Menu management
- Table management
- Order management (Dine-In and Takeaway)

- Order tracking system
- Bill printing
- Sales and profit reports
- Kitchen notification system

6. Database Tables

Table: staff

- id
- Name
- Cnic
- phone_no
- Position
- Salary

Table: restaurant_tables

- table_no

Table: orders

- order_no
- total_price
- is_paid
- order_date
- status
- table_no
- payment_status
- order_type
- status_time
- notification_shown

Table: order_items

- item_id
- order_no
- item_name
- quantity
- unit_price
- total_price

Table: order_counter

- id
- order_number

Table: menu

- id
- Name
- Category
- Cost_Price
- selling_Price
- image_path

7. System Screens (GUI)

Login Window – Allows multiple roles to access the system.

Admin Dashboard – Staff management, menu management, and table management.

Order Dashboard – Used by order taker to manage restaurant orders.

Takeaway Window – Used for takeaway orders.

Dine-In Window – Used for dine-in orders with table selection.

Order Tracking Window – Shows status of all active orders.

Owner Dashboard – Displays staff details, menu details, and sales reports.

Kitchen Dashboard – Displays orders and allows status updates.

8. System Architecture

Architecture Type: Two-Tier Client–Server Architecture

Client Layer: Qt Desktop Application developed using C++ and Qt Widgets.

Data Layer: MySQL database used to store restaurant information.

The client application directly communicates with the MySQL database to insert, update, and retrieve data related to orders, menu items, staff records, and sales information.

9. Data Flow Example

1. Staff selects items in Takeaway or Dine-In window.
2. Order is saved in the MySQL database.
3. Kitchen staff prepares the order.
4. Kitchen updates order status to 'Ready'.
5. Application checks database using QTimer periodically.
6. Ready notification appears on the order windows.

10. Conclusion

The Restaurant Management System improves efficiency in restaurant operations by replacing manual record keeping with a digital solution. It provides reliable order management, real-time kitchen tracking, and detailed sales analysis, helping restaurant management make better business decisions.